Math 110 — Week 1

Solutions

HW1 P1.

Idea. Each statement is about membership, so take one element and unfold the two definitions. For (c), ask which elements could be lost or gained.

1. (a) Let $y \in f(f^{-1}(V))$. Then $y = f(x)$ for some $x \in f^{-1}(V)$, and $x \in f^{-1}(V)$ means $f(x) \in V$. So $y \in V$.
2. (b) Let $x \in U$. Then $f(x) \in f(U)$, which is the definition of $x \in f^{-1}(f(U))$.
3. (c) Strict in (a): $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, $V = \{-1, 1\}$: $f^{-1}(V) = \{\pm 1\}$ and $f(f^{-1}(V)) = \{1\}$. Strict in (b): same f, $U = \{1\}$: $f^{-1}(f(U)) = \{\pm 1\}$.
4. Equality in (a) for a given V holds exactly when $V \subseteq f(X)$: an element $y \in V$ is in $f(f^{-1}(V))$ exactly when $y = f(x)$ for some x. It holds for every V exactly when f is surjective: if f is surjective, every $V \subseteq Y = f(X)$; conversely, equality for $V = Y$ gives $Y = f(f^{-1}(Y)) = f(X)$.
5. Equality in (b) for a given U holds exactly when no point outside U has the same value as a point of U. It holds for every U exactly when f is injective: if f is injective and $f(x) = f(u)$ with $u \in U$ then $x = u \in U$; conversely, if f is not injective, say $f(x) = f(x')$ with $x \neq x'$, then $U = \{x\}$ gives $x' \in f^{-1}(f(U))$ but $x' \notin U$, so equality fails for this U.

HW1 P2.

Idea. Compute the square, then one more product.

1. Let $\omega = (-1 + \sqrt{3}\,i)/2$. Then
$$\omega^2 = \frac{1 - 2\sqrt{3}i + 3i^2}{4} = \frac{-2 - 2\sqrt{3}i}{4} = \frac{-1 - \sqrt{3}i}{2}.$$
2. Multiply ω^2 by ω:
$$\omega^3 = \omega^2\omega = \frac{(-1 - \sqrt{3}i)(-1 + \sqrt{3}i)}{4} = \frac{1 - 3i^2}{4} = \frac{4}{4} = 1.$$

HW1 P3.

Idea. If $a \neq 0$ it has an inverse; multiplying by it isolates b. This needs $0 \cdot x = 0$, which is proved from the distributive law.

1. $0 \cdot x = (0+0)x = 0 \cdot x + 0 \cdot x$. Adding $-(0 \cdot x)$ to both sides and regrouping (inverse, associativity, inverse, identity):
$$0 = 0 \cdot x + \big(-(0 \cdot x)\big) = \big(0 \cdot x + 0 \cdot x\big) + \big(-(0 \cdot x)\big) = 0 \cdot x + 0 = 0 \cdot x.$$
2. If $ab = 0$ and $a \neq 0$, then a^{-1} exists (axiom (vii)), and by axioms (iv), (vii), (ii) and step 1,
$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

HW1 P4.

Idea. Same as P3, with a vector in place of b.

1. If $a = 0$ there is nothing to prove. Assume $a \neq 0$; then a^{-1} exists (field axiom (vii)).
2. By vector-space axioms (vii) and (v), the hypothesis $av = 0$, and Prop. 4(ii),

$$v = 1 \cdot v = (a^{-1}a)v = a^{-1}(av) = a^{-1} \cdot 0 = 0.$$

HW1 P5.

Idea. The zero function decides one direction; linearity of the integral gives the other.

1. If the set is a subspace it contains the zero function, whose integral is 0. So $b = 0$.
2. If $b = 0$: the zero function is in the set. If $\int f = \int g = 0$, then $\lambda f + g$ is continuous and $\int(\lambda f + g) = \lambda \int f + \int g = 0$.

HW1 P6.

Idea. A subspace of a complex vector space must be closed under multiplication by complex scalars.

1. A subspace of the complex vector space $\mathbb{C}^2$ must contain λv for every complex λ and every v in it.
2. $(1, 0) \in \mathbb{R}^2$, but $i \cdot (1, 0) = (i, 0) \notin \mathbb{R}^2$. So $\mathbb{R}^2$ is not a subspace of $\mathbb{C}^2$.
3. If $\mathbb{C}^2$ is regarded as a real vector space (only real scalars), then $\mathbb{R}^2$ is a subspace: it contains 0, and $\lambda v + w \in \mathbb{R}^2$ for real λ and $v, w \in \mathbb{R}^2$.

HW1 P7.

Idea. Every cross term carries a factor 2, and $2 = 0$ in $\mathbb{F}_2$.

1. $(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1$.
2. $(x + 1)^4 = (x^2 + 1)^2 = x^4 + 2x^2 + 1 = x^4 + 1$.
3. $(x + 1)^8 = (x^4 + 1)^2 = x^8 + 2x^4 + 1 = x^8 + 1$.
4. Over $\mathbb{R}$ the middle terms $2x$, $2x^2$, $2x^4$ do not vanish, and $(x + 1)^8$ has nine nonzero terms with coefficients $1, 8, 28, 56, 70, 56, 28, 8, 1$. Over $\mathbb{F}_2$ the seven middle coefficients $8, 28, 56, 70, 56, 28, 8$ are all even, so each of them is 0, and only $x^8 + 1$ remains.

R1.

Idea. Intersection: both conditions hold in each subspace. Union: closure under addition fails.

1. Let U_α be subspaces. 0 is in every U_α. If v, w are in every U_α, so is $\lambda v + w$. So the intersection is a subspace.
2. The union need not be: the two axes in $\mathbb{R}^2$ contain $(1, 0)$ and $(0, 1)$ but not their sum. See E1 for exactly when it is.

R2.

Idea. Existence: multiplying by the conjugate makes the product real. Uniqueness: put α between two candidate inverses and regroup.

1. Write $\alpha = a + bi$ with a, b real, not both 0, so $a^2 + b^2 > 0$. Let $\beta = (a - bi)/(a^2 + b^2)$. Then $\alpha\beta = (a^2 + b^2)/(a^2 + b^2) = 1$.
2. Suppose also $\alpha\beta' = 1$. Using associativity and commutativity of multiplication in $\mathbb{C}$,

$$\beta = \beta \cdot 1 = \beta(\alpha\beta') = (\beta\alpha)\beta' = 1 \cdot \beta' = \beta'.$$

E1.

Idea. If neither contains the other, pick one vector in each that is not in the other and add them; the sum cannot be in either.

1. If $U \subseteq W$ then $U \cup W = W$ is a subspace; likewise if $W \subseteq U$.
2. Suppose $U \cup W$ is a subspace and $U \not\subseteq W$. Pick $u \in U$ with $u \notin W$. Let $w \in W$ be arbitrary. Then $u + w \in U \cup W$.
3. If $u + w \in W$, then $u = (u + w) - w \in W$, which is false. So $u + w \in U$, and then $w = (u + w) - u \in U$. Hence $W \subseteq U$.

E2.

Idea. For two subspaces, direct means the intersection is $\{0\}$.

1. $U \cap W = \{(0, y, 0)\} \neq \{0\}$: not direct.
2. $U \cap X$: (t, t, t) with third coordinate 0 forces $t = 0$. Direct. Similarly $W \cap X = \{0\}$ (first coordinate 0): direct.
3. $U + X = \mathbb{R}^3$: every (a, b, c) is $(a - c, b - c, 0) + (c, c, c) \in U + X$, so $\mathbb{R}^3 \subseteq U + X$; and $U + X \subseteq \mathbb{R}^3$ because both are subspaces of $\mathbb{R}^3$.
4. The sum is all of $\mathbb{R}^3$ and $U \cap X = \{0\}$ (step 2), so $\mathbb{R}^3 = U \oplus X$ (Def. 6).